

Incident Response Report – Task 2 (SOC Internship Simulation)

1. Executive Summary

This assessment simulated a **Security Operations Center (SOC)** workflow using **Splunk** to ingest and analyze log data (`SOC_Task2_Sample_Logs.txt`). The goal was to detect suspicious events, classify incidents by severity, and provide remediation and communication steps.

The exercise demonstrates **SIEM alert monitoring, incident triage, and reporting**, aligned with industry SOC practices.

2. Environment Setup

- **Platform:** Splunk Free Edition (local instance on Kali Linux)
- **Dataset:** `SOC_Task2_Sample_Logs.txt` (includes login events, IP connections, malware detections)
- **Tools Used:**
 - Splunk (log ingestion, dashboards, queries)
 - Excel (`Alert_Classification.xlsx`) for severity categorization
 - Screenshots of Splunk dashboards

3. Findings

3.1 Suspicious Alerts Identified

From Splunk searches and dashboards:

1. **Multiple Failed Logins**
 - Several failed login attempts for users (`admin`, `wiener`) from unusual IPs.
 - Indicates brute-force attempts.
2. **Unusual IP Access**
 - Access attempts from IP ranges outside expected geography.
 - Could indicate reconnaissance or compromised credentials.
3. **Malware Detection Events**
 - Alerts showing malware signatures triggered on endpoints.
 - Requires immediate containment.
4. **Privilege Escalation Attempt**
 - Logs show suspicious `sudo` or `root` activity after failed logins.

4. Incident Classification (from `Alert_Classification.xlsx`)

- **High Severity**
 - Malware detection alerts
 - Multiple failed logins (possible brute force)
- **Medium Severity**
 - Unusual IP logins
- **Low Severity**
 - Single failed login attempts

5. Impact Assessment

- **Brute Force Attempts** → risk of account compromise.
- **Malware Alerts** → endpoint may be infected, leading to lateral movement.
- **Suspicious IP Access** → may indicate attacker probing systems.

6. Recommended Actions

- **Containment:**
 - Block malicious IP addresses at firewall.
 - Isolate infected hosts.
- **Eradication:**
 - Run antivirus/EDR scans.
 - Reset compromised user credentials.
- **Recovery:**
 - Monitor restored systems closely.
 - Harden SSH login (limit attempts, enable MFA).
- **Prevention:**
 - Enable Geo-IP filtering.
 - Implement SIEM alert playbooks for failed logins.

7. Dashboard Evidence

- Login attempts per user (bar chart).
- Failed logins per IP (bar chart).
- Malware alerts over time (timechart).
- Incident summary (pie chart).